SAEON User Stories and Tasks

Acceptance requirements and backlog links

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

These stories define the required behaviour. They do not prove that a feature has passed testing. Final field acceptance remains open. Final-sensor evidence is tracked under PB17; Firebase access checks under PB18; portfolio completion under PB19.

US01 Monitor Rainfall

Backlog links: PB01, PB02, PB10

As a SAEON researcher, I want the station to measure rainfall automatically so that I can collect observations without manually retrieving a USB data logger.

Acceptance criteria

- The TR-525M rain gauge produces a recorded count for each bucket tip.
- Contact bounce does not produce duplicate counts during controlled testing.
- The system converts counts into rainfall depth in millimetres using the verified calibration factor for the installed gauge.

Tasks

- Connect the rain gauge to the ESP32.
- Implement tip counting and debouncing.
- Implement rainfall calculations.
- Compare recorded results with controlled test measurements.

US02 Monitor Temperature and Humidity

Backlog links: PB01, PB02, PB10, PB17

As a SAEON researcher, I want temperature and humidity readings alongside rainfall data so that I can understand conditions at the station.

Acceptance criteria

- DHT11 is identified as the testing sensor and SHT31 as the final-product sensor in configuration and test records.
- The final SHT31 output includes temperature in degrees Celsius and relative humidity in percent.
- Temperature and humidity are checked against suitable references, with the comparison method and results recorded.
- Missing or invalid readings are flagged and are not presented as valid measurements.

Tasks

- Test DHT11 readings and error handling.
- Integrate the SHT31 configuration for the final product and record the actual verification results.
- Include temperature, humidity and sensor identity in the telemetry schema and test documentation.
- Compare readings with references and record the sensor used.

US03 Transmit Data Remotely

Backlog links: PB01, PB03, PB04, PB10

As a SAEON researcher, I want the station to transmit readings through a cellular network so that I can monitor remote sites without relying on local Wi-Fi.

Acceptance criteria

- The SIM7000G establishes a data connection using the configured SIM and network settings.
- Messages reach HiveMQ Cloud over cellular MQTT and the backend ingestion service validates and stores them in Firebase.
- Each message includes a station identifier, observation time and documented measurement fields.
- Connection failures are recorded and recovery behaviour is tested.

Tasks

- Configure SIM7000G wiring, power and cellular access.
- Configure MQTT topics and access permissions.
- Configure the backend ingestion service and its Firebase writes.
- Test publishing, reconnection and stored message consistency.

US04 Store Monitoring Records

Backlog links: PB01, PB03, PB04, PB18

As a SAEON researcher, I want readings saved in a database so that I can review environmental observations over time.

Acceptance criteria

- Valid incoming readings are stored in Firebase.
- Each reading is associated with the correct station and observation time.
- Invalid messages are rejected or recorded as errors.
- Stored records can be retrieved by the dashboard.

Tasks

- Define station and telemetry records in the chosen Firebase database service.
- Develop the MQTT-to-Firebase ingestion connection.
- Validate incoming messages and enforce authorised writes.
- Test storage and retrieval through the dashboard.

US05 View Current Conditions

Backlog links: PB05

As a SAEON researcher, I want a dashboard showing the latest rainfall, temperature and humidity readings so that I can check current conditions remotely.

Acceptance criteria

- The dashboard displays rainfall, temperature and humidity with their units.
- The station identifier and latest update time are visible.
- New readings update the displayed information.
- Missing or outdated data is clearly identified.

Tasks

- Create the dashboard layout.
- Connect the dashboard to monitoring data.
- Implement live updates.
- Add loading, error, and stale-data displays.

US06 Review Historical Data

Backlog links: PB05

As a SAEON researcher, I want to filter historical readings by station and date so that I can examine changes and compare observations.

Acceptance criteria

- Users can select a station and date range.
- Charts and tables show records matching the selected filters.
- Rainfall, temperature and humidity are labelled with the correct units.
- An empty result is clearly explained.

Tasks

- Implement historical data queries.
- Add station and date filters.
- Create rainfall, temperature and humidity charts.
- Test filters against stored records.

US07 Export Monitoring Data

Backlog links: PB07

As a SAEON researcher, I want to export selected readings as a CSV file so that I can analyse them using other tools.

Acceptance criteria

- The export matches the selected station and date range.
- Column headings identify measurements and units.
- Exported values match the records displayed in the dashboard.
- Timestamps use a documented time zone.

Tasks

- Create the CSV export function.
- Apply the dashboard filters to exported data.
- Add clear column headings.
- Compare an export with database records.

US08 Check Station Health

Backlog links: PB07

As a station operator, I want to view station health information so that I can identify power or communication problems.

Acceptance criteria

- The dashboard shows the latest available battery reading and last communication time.
- Missing health information is clearly identified.
- Low battery and stale communication are flagged using agreed thresholds.
- Old readings are not presented as evidence that the station is currently healthy.

Tasks

- Implement battery measurement and calibration.
- Include diagnostic fields in transmitted messages.
- Create the station health panel.
- Test low-power and connection-loss displays.

US09 Access the Dashboard Securely

Backlog links: PB03, PB05, PB18

As an authorised SAEON user, I want to sign in securely so that monitoring information and restricted functions are available according to my access permissions.

Acceptance criteria

- Valid users can sign in and sign out.
- Restricted pages require an authenticated session.
- Database access rules enforce the agreed permissions.
- Privileged database credentials are not exposed in the browser.

Tasks

- Configure Firebase Authentication and the agreed database access rules.
- Create the sign-in interface.
- Implement protected routes and session handling.
- Test authorised access and rejected unauthorised requests.

US10 Use the Dashboard on Different Devices

Backlog links: PB07

As a SAEON researcher working away from my desk, I want a dashboard that works on phones, tablets, and computers so that I can check station information wherever I am.

Acceptance criteria

- Key readings and controls remain readable on the tested screen sizes.
- Charts, filters, and navigation remain usable.
- Important information is not hidden by overlapping or clipped elements.

Tasks

- Implement responsive dashboard layouts.
- Adapt charts and tables for smaller screens.
- Test phone, tablet, and desktop layouts.
- Correct usability issues found during testing.

US11 Maintain Station Power

Backlog links: PB02, PB06, PB09, PB10

As a station operator, I want a suitable battery and charging arrangement so that the station can continue monitoring between maintenance visits.

Acceptance criteria

- Battery, charging, and solar components are checked for compatibility.
- Operating and sleep-mode power consumption are measured.
- Charging behaviour is verified.
- Operating duration is tested against the team's agreed requirement.

Tasks

- Assemble and test the power system.
- Integrate the solar charging arrangement.
- Implement and test low-power operation.
- Record power consumption and operating duration.

US12 Deploy and Validate the Station

Backlog links: PB08, PB09, PB10, PB13

As a SAEON researcher, I want the assembled station tested in its deployment environment so that I can assess whether its measurements and communication are reliable.

Acceptance criteria

- Electronics are secured in the prepared enclosure.
- Sensor placement and station installation are documented.
- A 48-hour field test records communication, power behaviour, and measurement issues.
- Rainfall, temperature and humidity readings are compared with reference measurements.
- Results, faults, and deployment limitations are documented.

Tasks

- Complete final assembly and installation.
- Perform baseline calibration.
- Start the 48-hour field test.
- Collect manual reference measurements.
- Audit the results and record required improvements.

US13 Maintain the System

Backlog links: PB01, PB11, PB12, PB13, PB19

As a future maintainer, I want clear technical documentation so that I can set up, troubleshoot, and maintain the station and dashboard.

Acceptance criteria

- Documentation covers components, wiring, configuration, and calibration.
- Firmware, cloud, database, and dashboard setup steps are included.
- Troubleshooting guidance and known limitations are recorded.
- Credentials are handled through documented configuration steps rather than published in the documentation.
- The setup guide states Firebase, DHT11 for testing and SHT31 for the final product consistently.

Tasks

- Complete wiring and power documentation.
- Write firmware, database, and dashboard setup instructions.
- Document testing and troubleshooting.
- Review the README files for clarity and completeness.

Recording acceptance

For each story, record the test date, test conditions, observed result, evidence reference and reviewer. Keep failed checks and limitations visible. Firebase database service selection belongs in the implementation guide; no specific database product or deployment configuration is assumed here.

Requirement reference: PRP37XS Subject Guide, pages 8 and 20.